

Water networks contribution to Demand Response

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Abstract—Energy costs is a challenge for water distribution networks companies. In the same time, the smart grid provides new opportunities for cost optimization. This paper demonstrates how to use the intrinsic flexibility of water distribution networks (reservoirs and pumping plans), to offer energy modulation services while respecting operational and committed quality of service.

Index Terms—Demand response for water network, technical aggregation

I. INTRODUCTION

The virtual energy market is taking place. New actors are collecting opportunities to reduce subsequently energy demand for short periods of time at peak hours, or at the opposite to absorb energy surplus in case of high production level (in particular from renewable sources) at low demand hours. The value of these new services to the grid is very variable, depending on the energy market dynamic prices.

Considering the large volume of water storage and the pumping capabilities, the water networks offer good opportunities for this business. To make this real, it is necessary to provide new control systems combined with analytic decision making tools, ensuring that the water demand will be entirely fulfilled, evaluating the economic equation, and providing the best strategy to make the most of the virtual energy market.

II. THE CONTEXT OF WATER NETWORKS

Energy for water industry represents 3 % of world energy consumption. The total electricity consumption of the water and wastewater sectors will grow globally by a predicted 33% in the next 20 years especially due to the increasing number of people getting drinking water (world population increase combined to development of the infrastructure).

On the operation side, 1/3 of total operation cost is energy, mostly electricity for pumping [1]

Due to regulated water prices, increase of energy prices and commitment to promote water demand reduction, business context is under stress and water companies need to preserve their profit by increasing efficiency and especially energy efficiency.

On the technical side, a new trend has emerged in water distribution networks: with the increased level of instrumentation and telemetry, a new layer of smart data applications has become possible. “Smart Water Network” provides better measuring, collecting, analyzing, and acting upon a wide range of network events, improving the efficiency, longevity, and reliability of the underlying physical water network and makes possible new high level services based on the leverage of this information.

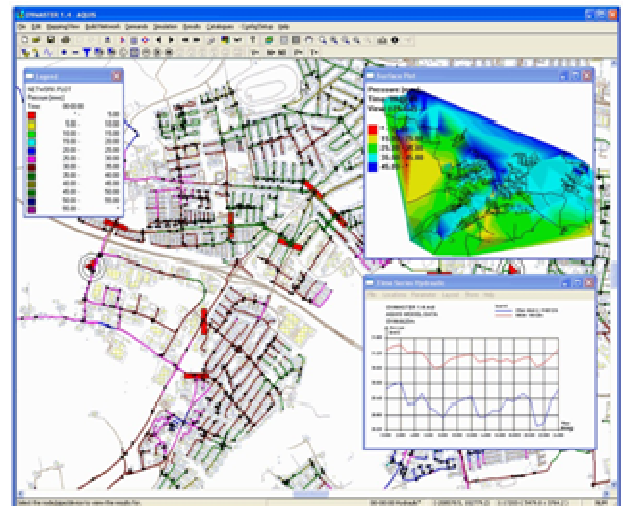


Figure 1: Hydraulic real-time operators interface

III. THE EXPERIENCE OF BIRKEROD SIMULATED WATER NETWORK

As a global specialist in energy management with operations in more than 100 countries, Schneider Electric offers integrated solutions to make energy safe, reliable, efficient, productive and green across multiple market segments. All over the world, electricity networks are becoming more complex and less stable. To continue to efficiently balance supply & demand, the grid needs to become smarter: The Grid will be Smart once ax²ll users are smart grid-ready and connected to the network.

To feed that vision, Schneider Electric Corporate Research is developing -and testing through pilot applications- new features that enable demand response mechanism, while preserving end users satisfaction, and being compatible with current practices.

A. The demonstration project

The project which is presented here illustrates the decision making tool and the technical aggregation for participating to virtual energy market in the case of a water network. This demonstration is based on a simulated environment which is extracted from the distribution network of Birkerød a small town located in Denmark. The goals of the demonstration are to implement and demonstrate a prototype of the tool in the context of a simulated Water Distribution Network, including the energy management system. A special attention was paid to make sure that operational conditions are taken into account and presented in a usable format to the user.

B. Use case

In order to be able to demonstrate our technical approach we needed to make some hypothesis on a commercial environment, as this market is not completely settled yet and is currently different from one country to another.

We have considered a situation where the Water Supply Utility is connected to an electricity aggregator who is trading energy and power on the market. We do not detail the type of contract, and focus on the information exchanges needed for this type of commercial interaction.

The water supply utility, as a customer of the electricity aggregator, provides the information of its energy consumption modulation capability. This information is given for the day ahead as a list of curtailment or overconsumption capability described on an hourly basis. The electricity aggregator selects the offer that best suits its aggregation strategy. The water supply utility confirms its commitment to the modulation (see figure 2)

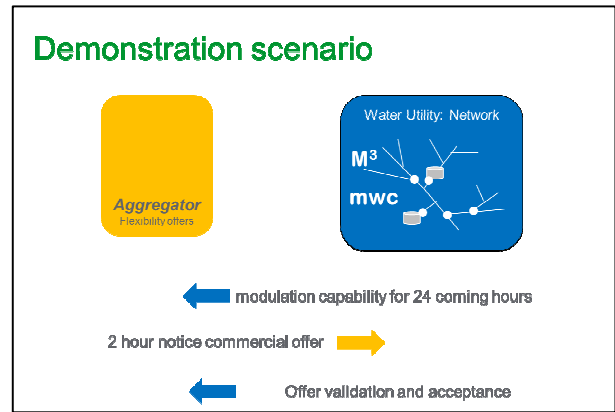


Figure 2: Description of the use case

C. Modulation calculation

The operator is managing a pumping plan to deliver the right level of pressure to the customers

There are operational constraints to be respected (minimum level of pressure for the fire brigade, pressure and pressure variations limits)

D. Operator interface

The interface is centred on the operational management of the water network. Even if there is the objective to optimize

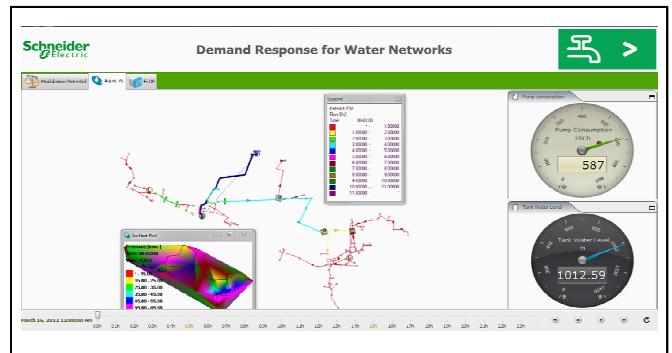


Figure 3: Interface showing the water network flows and pressure levels (hydraulic view)

the energy consumption for the management of the drinking water distribution network, we do not forget that the first and foremost goal is to provide the right water pressure at the consumption points, and to comply with the operational constraints of the water network.

On the example we propose three tabs for three different types of information.

1) Hydraulic view

The first one is the hydraulic view of the network which is a very classical hydraulic network view. It shows also some gauge that indicates the water level in the water tower and the current power used by the system

2) Demand Response view

This view is the dashboard of the Demand Response system whose mission is to identify from the established pumping plan, what can be modified to generate consumption modulation while respecting the operational goals and constraints.

On this view are presented several graphs. The first graph is the result of the computed optimization; it presents all modulation possibilities on an hourly basis. A red bar means that the corresponding power can be curtailed for one hour, a blue one means that the corresponding power can be used for one hour.

In the use case, one action only is selected. It will be performed if agreed by both the aggregator and the water network operator.

The second graph shows the pumping plan and the modification proposed by the system. It is linked to the previous one and presents the updated pumping plan corresponding to the selected modulation offer. On the screen hard copy, one can see the alteration of the pumping plan corresponding to the curtailment at 01:00 in the morning (arrow 1 in the enlarged picture) and the effect on the plan at 15:00 and 17:00 (Arrow 2). The additional pumping is mandatory to keep the water tank at the right level. Let's note that the optimization system take into account the overall cost of the modification and propose only modifications which have a net benefit on the operational costs.

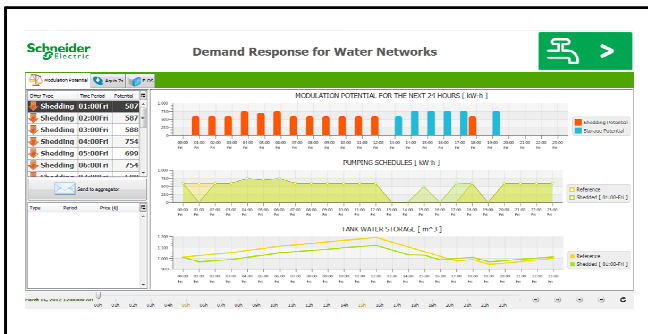


Figure 4: Demand Response dashboard

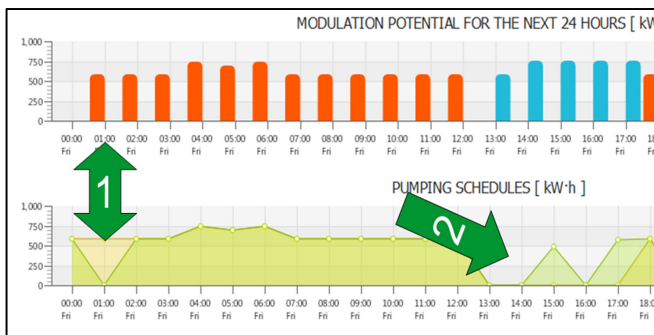


Figure 5: Modification of the pumping plan corresponding to the 01:00 curtailment

E. Architecture of the system

The architecture reflects the logical approach which we have taken. The optimization engine is based on two forecast input data:

- Energy price
- Water demand profile

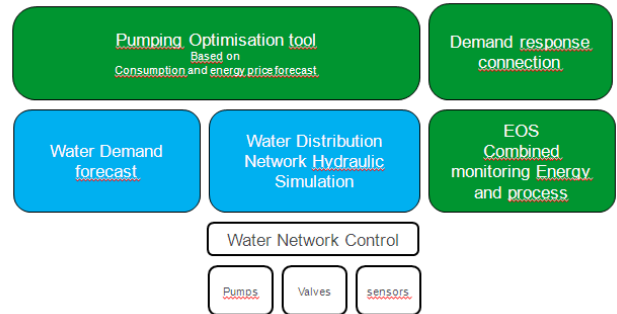


Figure 6: Functional Architecture

3) Optimisation tool

The optimization tool uses the water distribution network hydraulic model with friction loss developed by [2]. This model allows minimizing pumping costs according to a fixed energy supply tariff. The originality of the proposed model is to incorporate shedding operations.

Friction losses in pipes are calculated by the formula of Darcy-Weisbach for turbulent flow. The dimensionless friction coefficient λ was chosen with the simplified law of Prandtl-Karman for hydraulically rough.

The non-convexities induced by the head loss equations, the yields of the pumps and the mandatory Integer variables (0-1) used to model the starting and the stopping of the pumps make the resulting problem particularly challenging to solve to optimality.

Hence an heuristic first pass obtained via linearization of the MINLP model was used to strengthen the robustness of the results and to guide the solver to "promising" regions of the search space.

First, a reference pumping plan is established by letting the optimizer model choose the best decisions considering both the market conditions and the water distribution network operational constraints (minimum head needed).

Then and for each time period (typically an half-hour period is considered) of the simulation time window, one optimization is launched with the added constraint of either to shed the totality of the energy for this specific time period and to shift it on consecutive time periods in an optimal way, or to over-consume at this specific time period, thus giving the ability to reduce the energy consumption later.

This process yields two pumping plans per time period (one for shedding hypothesis, one for the over consumption one) with the corresponding financial gain or loss computed by the optimizer. The pumping plans are then finely simulated with the hydraulic simulation tool described thereafter.

The AMPL modeling language was used for stating the problem in a high level mathematical language in conjunction with the Knitro© non-linear solver from Ziena Inc [3] as the Mixed Non Linear Program solver. Artelys Risk Manager™ [4] was used for the development of the WDN optimization models and for the study.

4) Hydraulic simulation tool

The hydraulic simulation tool is based on AQUIS tool. Aquis is a hydraulic modeling tool which enables simulation of flows and pressure everywhere in the water network. One major strength of this tool is its ability to be connected to real time measures through SCADA operation tool. Then, it is possible to calculate the future evolution of the network (in terms of flow, pressure, tanks level, energy consumption, ...) for different scenarios of pumping scheduling, taking into account various water demand from the users side.

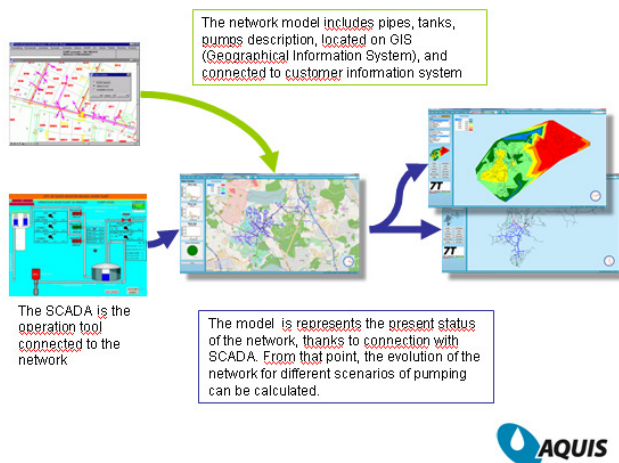


Figure 7: Overview of Aquis, hydraulic simulation tool

5) Water demand forecast and demand response connection

As the demonstration was focused on the optimization principles, no forecast algorithm was implemented and water demand was given as input.

IV. RESULTS AND PERSPECTIVES

This demo showed the potential of modulation in a realistic water network. Savings expectations are real: 5 to 10 % for a plan that takes variable energy price at D-1, 10 to 15 % for intraday capacity market contract, depending on countries, electrical network characteristics and energy market maturity.

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